

WASP-11b

R-0905

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_240924 · created 2026-07-29T04:34:16+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460577.9288 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.086 +/- 0.041
Transit depth (Rp/Rs)^2	0.74 +/- 0.71 [%]
Orbital Inclination (inc)	89.17 +/- 1.86
Airmass coefficient 1 (a1)	1.119 +/- 0.014
Airmass coefficient 2 (a2)	-0.11 +/- 0.082
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	#3 - [458, 373]
Optimal Aperture	4.5
Optimal Annulus	7.5
Transit Duration (day)	0.094 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.086 $\pm$ 0.041 vs 0.131 (deviation 1.1 σ)
Duration fit vs published	0.094 d vs 0.1075 d (deviation 0.84 σ)
Mid-time O-C	9.3 min
Depth SNR	2.1
Residual scatter	1.22 %

Light curve

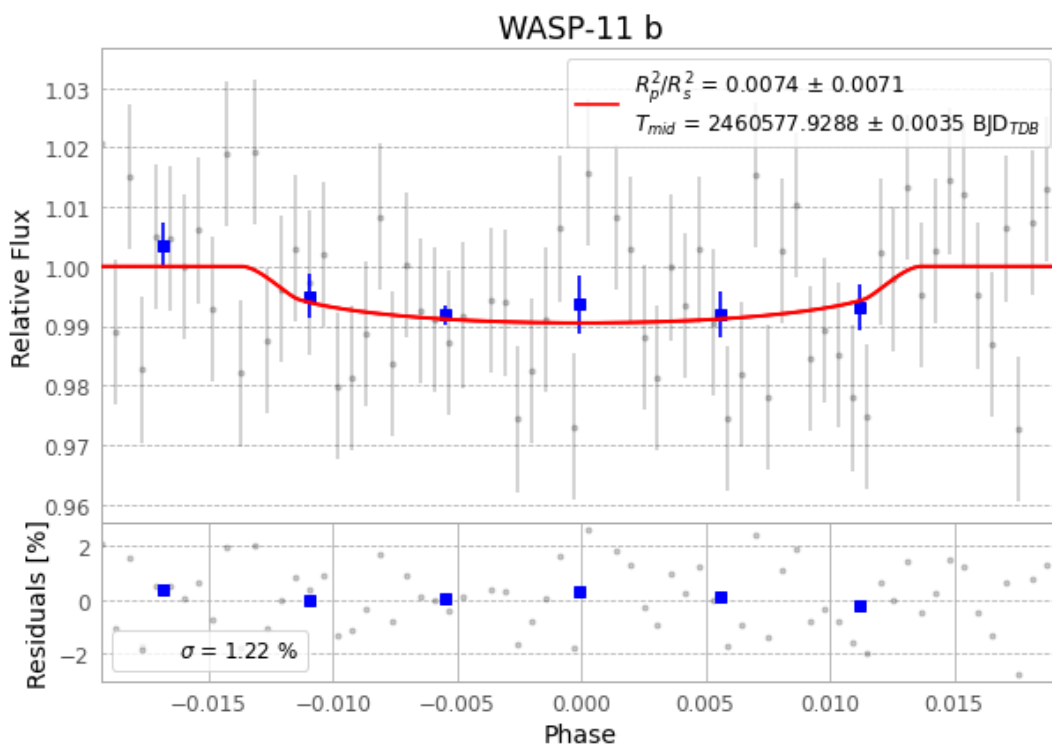


Figure 1 — FinalLightCurve_WASP-11 b_2024-09-24.png (SHA-256 abbad8d7176f0cdc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [405, 338], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c3b7a4506ac81f83...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

69 FITS frames (45 MB), HATP-10240924082715.FITS → HATP-10240924115416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-240924.log (5df6355061e2...), FinalLightCurve_WASP-11 b_2024-09-24.png (abbad8d7176f...), FinalLightCurve_WASP-11 b_2024-09-24.csv (33b72a9358b9...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 04:34Z.